Mid Ques Solve

True / False -

1. When the microprocessor has an operating system, the user must work directly with physical I/O devices.

   => False

2. 8086 microprocessor can map upto 10 MB of memory.

   => False

3. In 8086, most arithmetic and logical operations are performed in Ex register.

   => False

4. 8086 has 6 segment register.

   => False

5. ~~TES~~ TEST pin is examined by the READY instruction.

   => False

6. Cycle transfer is one type of DMA.

   => True.

Meaning -

DDR = Data Direction Register

Used to configure I/O port pins as input/output.

INTR = Interrupt Request

A maskable interrupt input pin used for completing current instruction & interrupt acknowledge operation

TRAP - A ~~Non Maskable~~ Internal Interrupt.

IP = Instruction Pointer

Holds the offset address of the next instruction to be fetched.

PC = Program Counter

Holds the address of the next instruction.

Conditional & Unconditional Programmed I/O.

=>

Unconditional I/O -

The microprocessor doesn't check wheather the I/O device is ready or not. It directly executes I/O commands and waits untill it completes.

Ex -

```
OUT 20H, AL
    PORT 20H
```

Here microprocessor directly outputs the data to the port without checking if the device is ready or not.

Conditional I/O -

The microprocessor checks the status of the I/O device is ready or not. If the device is ready, then data transfer occurs.

Ex -

```
CHECK : IN AL, 21H
        TEST AL, 01H
        JZ CHECK
        IN AL, 20H
```

Here, CPU first check if the device is ready before sending data.

Compare Memory Mapped I/O with Standard I/O according to their performance -

=>

Standard I/O -

For an 8 bit microprocessor, an 8-bit address is used for each I/O port, 16-bit address for memory location.

Microprocessor outputs high on IO/M' control pin to indicate that an I/O operation is taking place.

IN AL, PORTA and OUT PORTA, AL are the example of Standard I/O.

Memory Mapped I/O -

MSB of the address is used to distinguish between memory and I/O. If 1 then I/O, if 0 the memory.

Reduce microprocessor addressing memory by 50%.

MOV AL, START and MOV START, AL are the example of Memory Mapped I/O.

Special functions of General Purpose Register.

⇒

8086 has 4 general-purpose register - AX, BX, CX, DX.

AX -

Accumulator Register (AX) is used in arithmetic, logic and data transfer instructions, as it generates shortest machine language code. Must be used in multiplication, division, I/O operations.

BX -

Base Register (BX) used as a address register. Also used in array operations and table lookup operations.

CX -

Count Register (CX) used as loop counter. Used in shift and rotate counter.

DX -

Data Register (DX) used in multiplication and division. Also used in I/O operation.

How does a 8086 computes the 20 bit physical address from different 16 bit segments?

=>

8086 has 16-bit registers but 20-bit address bus. Therefore, it forms the physical address using a segment register and an offset.

Code Segment -

CS : IP = 0400H : 0056H

∴ Physical address = 04000H
+ 0056
04056H

Data Segment-

DS : EA = 0500H : 0920H

∴ Physical address = 05000H
+ 0920H
05920H

Stack Segment-

SS : SP = 0A00H : 0100H

∴ Physical address = 0A000
+ 0100
0A100H

Logical Address vs Physical address -

=>

A logical address is the address written in segment-offset form. ~~But a~~ Segment and offset both are 16 bit. But a physical address is the actual 20-bit address placed on the address bus.

Logical address = Segment : Offset

= 1230H : 0045H

Physical address = 12300

$\oplus$ 0045

12345H

↳ Absolute / Physical address.

Explain different flags available in flag register.

=>

In 8086, there are 9 flags in flag register.

Carry Flag (CF) - When unsigned carry occurs.

Parity Flag (PF) - Set to 1 when result has even number of 1's.

Auxilary Flag (AF) - Set to 1 when unsigned overflow.

Zero Flag (ZF) - Set to 1 when result is 0.

Sign Flag (SF) - Set to 1 when result is negative.

Overflow Flag (OF) - Set to 1 when signed overflow.

Trap Flag (TF) - Used for single step execution.

Interrupt Enable Flag (IF) -

If IF = 1, Processor responds to muskable interrupt.

Direction Flag (DF)-

DF = 0 ; Forward direction

DF = 1 ; Backward direction.

What is flag register ?

=>

Flag register is a 16 bit register that shows the status of the processor after instruction execution.

~~# Does flag register~~

Does 8086 use different flags to control program status?

=>

Yes, I agree.

All 9 type of flag register description as previous ques.

// Add this part-

ADD AL, BL

If result is zero, ZF = 1

If carry occurs, CF = 1

If signed overflow, OF = 1

If result is negative, SF = 1

Interrupts -

=>

There are mainly 3 types of interrupts -

1. Predefined (0-4)
2. Reserved (5-31)
3. User defined muskable interrupt (32-255)

Predefined interrupts -

Type 0	Divided by zero	- 0002H : 0000H
Type 1	Single step	- 0006H : 0004H
Type 2	NMI	- 000AH : 0008H
Type 3	Break point	- 000EH : 000CH
Type 4	Overflow	- 0012H : 0010H

Suppose during execution a microprocessor faces a power failure. Explain how it will respond to. Which pins shall be functional in this operation?

=>

A power failure is normally hadled using NMI. pin. Non-Muskable Interrupt (NMI) has higher priority than Muskable interrupt request pin (INTR). When a power failure is detected, a power monitoring circuit sends a signal to the NMI pin. Receiving the NMI signal, 8086 immediately suspends the current program causing Type 2

interrupt. After executing ISR the system completely shuts down.

What is pipelining? How does pipelining enhance computation speed of 8086?

=>

Pipelining is a technique where the 8086 microprocessor works on multiple instructions at the same time to improve speed.

Lets see how pipelining enhances the computation speed of 8086-

The 8086 has 6 byte instruction queue allowing it to fetch the next instruction while executing the current one.

Without pipelining-

Fetch instruction → Execute instruction → Fetch next instruction → Execute next instruction

With pipelining-

BIU fetches next instruction while EU executes current instruction.

DMA and How it ~~performes~~ enhance performance?

=>

Direct Memory Access (DMA) is a technique that transfers data between a memory and I/O device without involving the microprocessor.

This direct data transfer helps the microprocessor to spent less time in managing data movement and more time in computing. This is how DMA enhance the performance of microprocessor.

Draw the interrupt structure in 8086.

=>

Interrupt Type Code		20 bit Memory Access	
0	IP	00001H	00000H
	CS	00003H	00002H
1	IP	00005H	00004H
	CS	00007H	00006H
2	IP	00009H	00008H
	CS	0000BH	0000AH
⋮	⋮	⋮	⋮
255	IP	003FDH	003FCH
	CS	003FFH	003FEH

#. 8086 microprocessor operates in both 8085 and 8086 modes. Do you agree?
Discuss how 8086 microprocessor deals with 8085 environments.

=>

Yes, I partially agree.

The 8086 microprocessor operates in 2 modes -

Minimum Mode (equivalent to 8085 environment)

Maximum Mode (true 8086 mode)

The 8086 microprocessor deals with the 8085 environment in Minimum Mode ($MN/\overline{MX} = 1$), It directly generates all bus control signals like $\overline{RD}$, $\overline{WR}$, $M/\overline{IO}$ and $\overline{INTA}$ just like 8085. It uses a multiplexed AD0 - AD15 bus with an ALE signal to separate address and data, making it compatiable with 8085.

How partial decoding is better than linear decoding in memory design.

=>

Partial decoding is better than linear decoding because it uses decoding logic to select memory devics more efficiently. Partial decoding provides better address organization, reduce address space wastage and offers more reliable chip selection making it suitable for longer memory systems.

Suppose you want to generate an interrupt in 8086 mode. Explain the scenario with the envolvement of direct pins & signals.

=> Explaination-

1. First of all, A hardware device sends an INTR signal to the CPU.
2. If IF = 1, then the CPU sends INTA and receives the interrupt type number.
3. CPU saves current state. Then automatically pushes Flags, CS, IP onto the stackand jumps to ISR.
4. ISR ends with restoring Flags, Cs and IP.

#Demonstrate 8086 maximum mode briefly.

=>

In maximum mode, the 8086 is used in a multiprocessor system. The 8086 enters maximum mode when MN/$\overline{MX}$ ~~Pin~~ is 0. The processor derives the status signal S2, S1, S0. Another chip called bus controller derives the control signal using this status information. In maximum mode the S2, S1, S0 pins makes different combinations for different CPU cycle. In this mode,

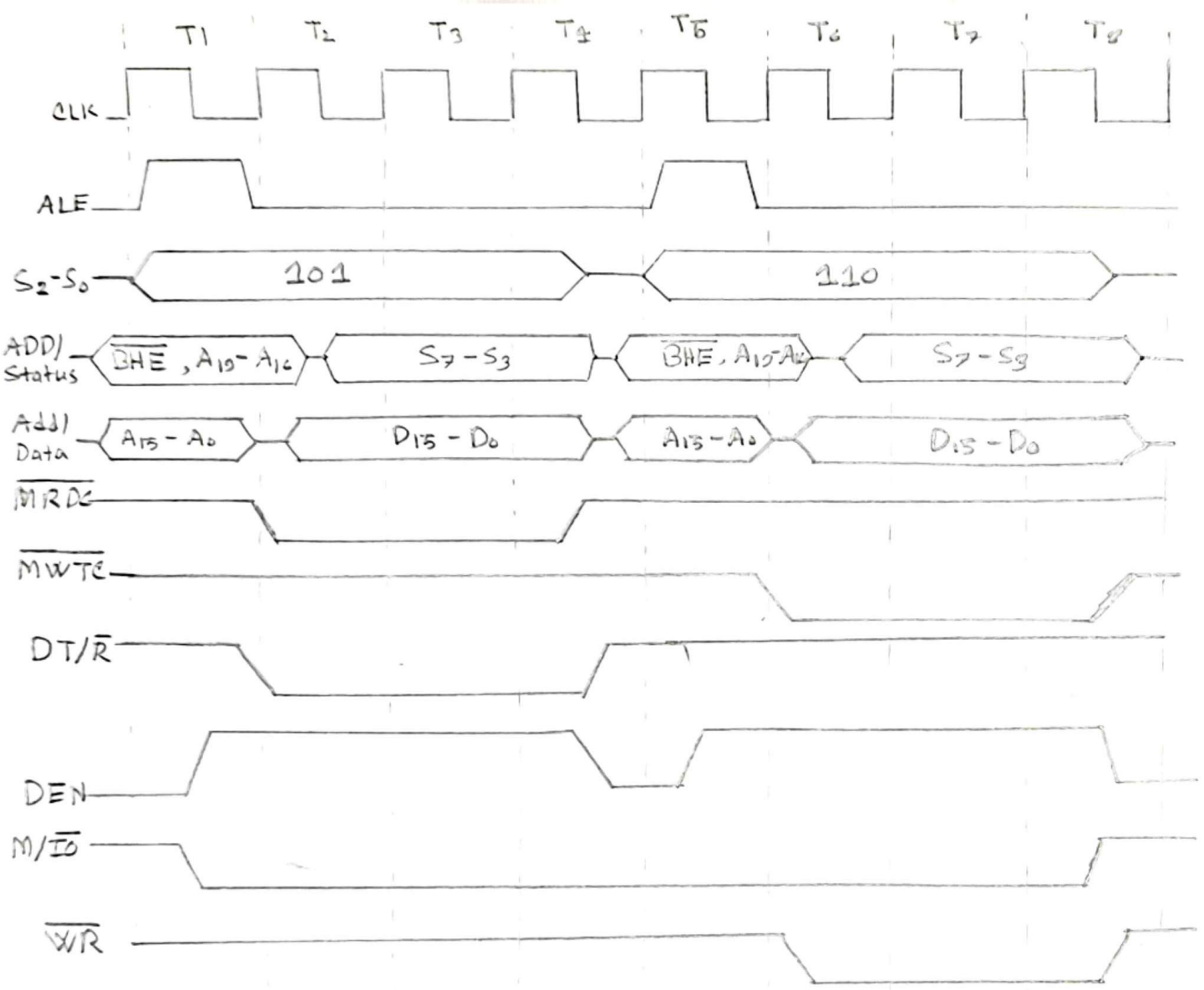
MRDC = Memory Read

MWDC = Memory Write

IORC = I/O Read

IOWC = I/O Write.

Suppose you have 2 instructions to execute.
Instruction 1 = S2 S1 S0 pins will be 101.
Instruction 2 = S2 S1 S0 " " 110.
Draw the machine cycle according the instruction & describe.



Description–

In maximum mode, S2 S1 S0 decides the bus operation. Here 101 = Memory read, 110 = Memory write.

For 101, the $\overline{MRDC}$ activates and memory sends data D15-D0. So $DT/\overline{R} = 0$. For 110, $\overline{MWTC}$ activates, so $DT/\overline{R} = 1$.

$M/\overline{IO}$ stays low for memory operation. $\overline{WR}$ low during memory writ